RevTrace: Preserving Valid Progress across Trusted Feedback Revisions

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***Abstract*—Feedback-driven inference can retract timely output whose computation remains valid. RevTrace coordinates three decisions: complete-input eligibility, original selection authority, and support for the Consumer's whole hypothesis. It retains eligible calls without endorsing their predictions and preserves accepted results across producer failure. We give sufficient retention conditions and a conditional selected-trace refinement argument. In a predeclared 24-speaker stratum, computation retention increases frontier withdrawals from 12 to 17; joint retention yields zero. In a matched four-speaker native correction study, offered audio coverage meeting both delivery deadlines rises from 45.2% to 99.2% with unchanged selected histories. Whole-run passes increase from zero with conservative execution to six of eight with RevTrace. No illegal installation occurs among 3,976 audited transitions in fault and sustained-state tests; cold recovery and no-reset quality failures delimit the service claim.**

***Index Terms*—dependable inference, feedback state, incremental computation, speculative execution, streaming ASR**

I. Introduction

A streaming service can meet its delivery deadlines and still erase usable progress. A correction selected now can become the input to later recognition [1]. This feedback dependency creates a service problem beyond first-delivery latency. In an actual streaming run, a complete two-second audio interval was visible 1.44 s after it became ready, meeting its two-second display deadline (D2). At 1.54 s, a trusted update replaced the whole hypothesis with a shorter snapshot despite its still-valid dependency path. Identical text returned at 3.72 s, meeting the eight-second certified-delivery deadline (C8). Both first deliveries were timely, yet the interval disappeared for 2.176 s (Figure 1). We call this an *avoidable withdrawal*: an update retracts a delivered whole view that it still supports. The client loses progress that the revision did not refute.

The separate RevTrace run in Figure 1(b) first displays the same whole text slightly after D2, then retains coverage. Its own next selected update also keeps an ahead view. The trajectories expose first delivery and continued availability as separate obligations.

A dependable revision service must both reject invalid results and preserve progress that remains usable. Discarding all older work enforces a conservative boundary at the cost of service. Preserving computation alone leaves another gap: the Consumer can still replace its valid ahead view when a selected state installs. In a predeclared 24-speaker stratum, computation retention improves timely audio coverage while increasing backward movements of the displayed audio frontier, or *frontier withdrawals*, from 12 to 17. Coordinating computation and view retention yields zero. Valid execution must therefore be connected to continued availability at the actual Consumer.

One feedback revision can have different consequences for an executing call, an accepted result, and a displayed hypothesis. RevTrace separates three judgments. **[EL] Input eligibility** checks the complete frozen input and its dependency path under the selected feedback state. **[AU] Selection authority** keeps the original selection and ordered installation bound to the complete original result. **[SUP] View support** checks the Consumer's whole hypothesis when that state installs (Figure 2). A call's input can be confirmed even when its own next prediction is wrong. A view can lose support while a future call's input is unresolved. Producer exit can end new-message authority while leaving an already accepted result eligible. A birth version, matching text, or producer-liveness bit cannot stand in for these different decisions.

Dependency reuse and revocation have established foundations in incremental dialogue and computation [2]–[4]. SSBD coordinates draft reuse with display masking [5]; PredGen speculates across evolving speech stages [6]. RevTrace realizes coordination around a recursively selected feedback task: complete calls retain the original correction and update, while the actual whole view is checked independently at selected-state installation. This provides a concrete use for old work without turning a prediction into selection authority.

We make three contributions:

- A service contract exposing avoidable withdrawal and sufficient, executable witnesses for complete-call eligibility and whole-view support across trusted revisions (Sections II–III).
- A joint coordinator that preserves the original selected transition relation while retaining eligible calls and supported whole views. Its conditional selected-trace argument connects early execution, durable selection and supported process failures (Sections III–IV).
- Evidence that computation-only retention can worsen visible progress, and that confirmed full calls improve timely original-task service. Factorial, native and fault controls distinguish these benefits from text accuracy, recovery latency and load boundaries (Section V).

In the matched four-speaker native study, audio coverage meeting both deadlines increases from 45.2% with conservative execution to 99.2% with RevTrace; whole-run passes rise from zero to six of eight. A whole-run pass meets both P99 delivery targets. The independent factorial study explains why retaining work and retaining its delivered view are separate obligations.

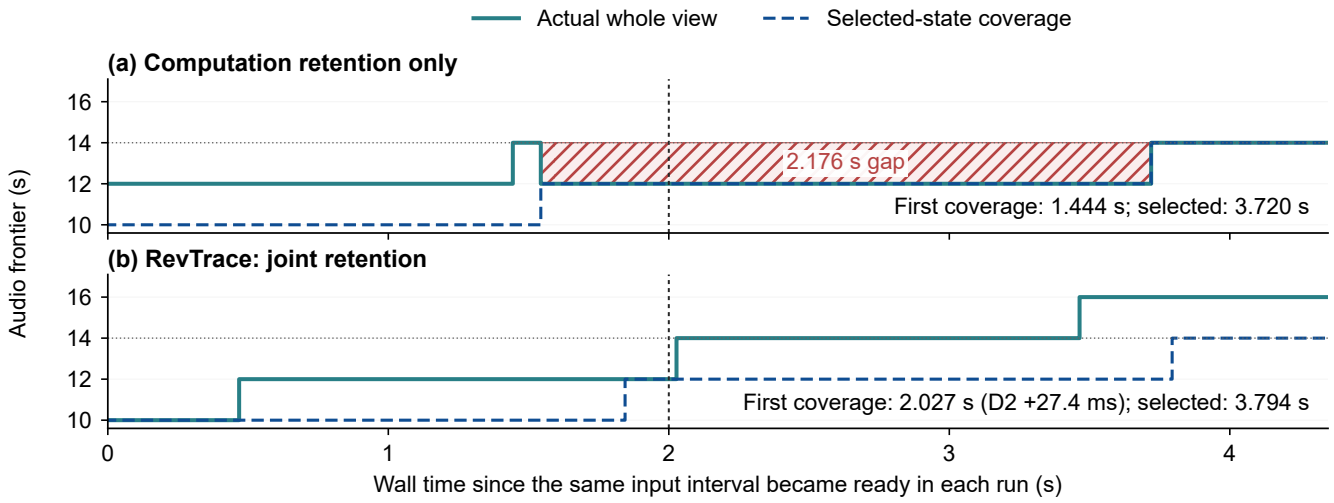


Fig. 1. On-time delivery does not guarantee continued availability. Two actual c16-overload runs process the same complete 12–14-s audio interval using their own ready clocks. (a) Computation-only displays 37 new code points at 1.444 s, withdraws the supported view at 1.544 s and restores identical text at 3.720 s. (b) RevTrace first displays the same whole text at 2.027 s, missing D2 by 27.4 ms, and keeps the interval covered thereafter. Its 3.794-s selected update retains an ahead 16-s view. The dotted line marks D2 (2 s); both certified deliveries meet C8 (8 s). These are separate executions with different interleavings, rather than an event-matched counterfactual.

II. System and Feedback Contract

A. System and Failure Model

Each session has an Owner serializing selected transitions, workers executing frozen calls, and an actual Consumer applying selected facts and complete provisional hypotheses. Durable storage records the selected prefix. A Supervisor monitors the processes and preserves the accepted-input ledger.

We assume one trusted Owner per session, protected selection, immutable audio and model configuration, unambiguous encoding, collision-resistant digests, and storage honoring successful synchronization. Used calls must faithfully consume their frozen inputs. Host receipts bind API objects, not GPU arithmetic; independent Private and Authority floating-point executions need not agree.

The interface checks identity, completeness, derived state, and ancestry. Native resilience supports worker or Owner-group fail-stop on an intact filesystem; Supervisor, accepted-input ledger, actual Consumer, power observer, and synchronized storage survive. Their original input clocks continue through restart. Supervisor/Consumer failure, host power loss, compromised Owner or Authority, Byzantine code, persistent corruption, and lost storage are excluded. Controlled wrong-input injections in Section V intentionally violate normal Private input fidelity; we use them to study containment and executor release, not as examples covered by the lawful-continuation proposition.

B. Feedback creates an execution dependency

An immutable audio stream supplies a protected selection path producing *Canonical* facts and a faster *Private* path computing and displaying provisional hypotheses. Selection can change the next call's input: StreamCorrect feeds its chosen correction into subsequent ASR [1]. We coordinate this dependency as the paths advance asynchronously.

For a fixed session and execution contract, write an installed fact as

$$C = (v, F, Z_C, P_C, T_C). \tag{1}$$

Here v is the version, F the certified processing frontier, Z_C the complete selected output, P_C the feedback state, and T_C its bound text. The contract fixes the model, adapter, decoding, audio identity, and transformations to P and T. Consumer coverage follows the adapter's publication policy; advancing F need not add characters. Certification means acceptance by protected selection and installation, separately from human-transcription accuracy.

The adapter declares a conservative representation of *all mutable cross-call state that can affect later model inputs, selection, or feedback*. Telemetry and resources are separate. Qwen passes prompt tokens without cross-call numerical KV caches. The native adapter additionally carries audio buffers, token groups, context, alignment position, and decoder configuration. Comparisons require the same contract: text equality cannot replace an omitted cursor or numerical cache. The representation may conservatively include fields unnecessary to a particular call.

C. Calls, views and outstanding input

A complete provisional node n records a frozen call

$$q_n = (a_n, P_n^-, b_n, \chi_n), \quad n = (q_n, Z_n, P_n^+, T_n), \tag{2}$$

where a_n is the input-state frontier, $b_n > a_n$ the audio target, and χ_n binds remaining inputs and execution identity. The model may read the prefix through b_n; a_n does not assert token-to-audio alignment. The adapter derives P_n^+ and T_n from a complete result. An in-flight call has only its frozen input; partial or cancelled tokens supply no node.

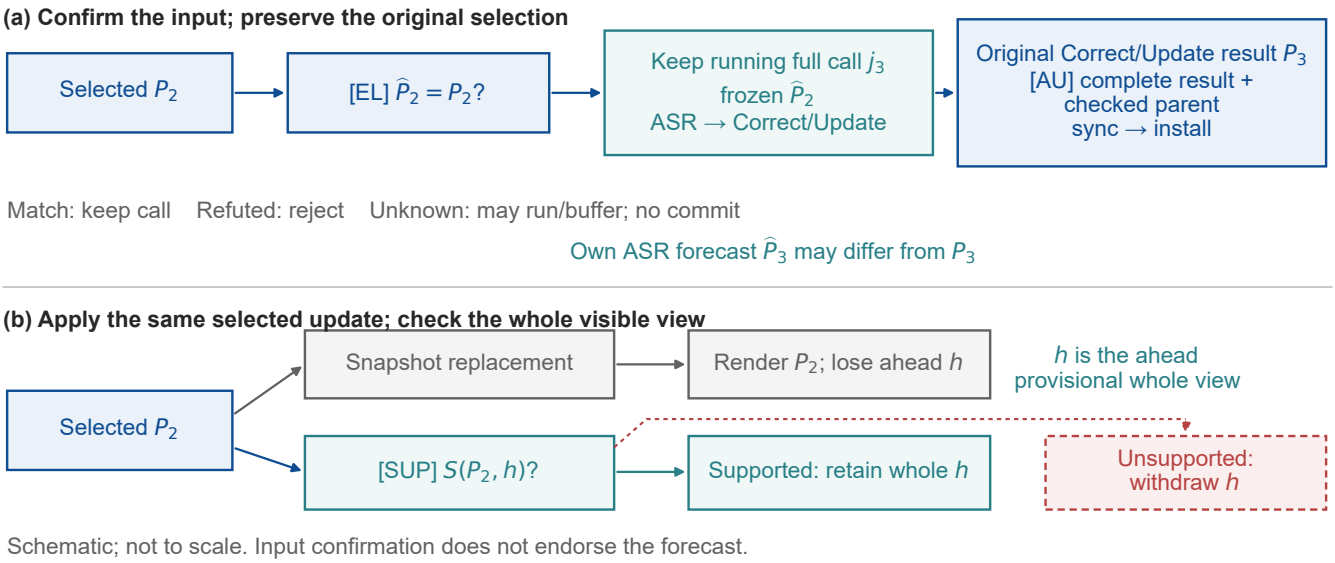


Fig. 2. One trusted revision requires separate decisions about work and its use. [EL] confirms the complete frozen input of j_3 against actual selected P_2. [AU] retains original Correct/Update before durable selection; the call's own prediction can differ from its result. [SUP] independently retains the whole supported view when P_2 installs. Unknown input remains unresolved while owned calls may run or buffer; it grants no commit authority. Arrows show decisions and their outcomes; the dashed red branch marks failed support. This is a causal schematic without a measured time scale.

The *dependency lifetime* determines usability under the current root; the *visible lifetime* determines whole-hypothesis retention; the *physical lifetime* ends only when execution finishes or drains. A new root can change the first two before the third ends. Their decisions therefore require separate evidence.

The service retains every accepted input obligation until certified processing covers it. Discarding a branch, requesting cancellation, or recovering an old view does not remove that obligation or reset its arrival clock. In exact mode, a missed deadline never authorizes an uncertified Canonical transition.

III. Coordinating Calls and Whole Views

A. One revision, three different decisions

Figure 2 follows a complete call j_3. Recognition in its parent produces a provisional state $\widehat{P}_2$ while original Correct/Update is still selecting P_2. RevTrace can start j_3 on $\widehat{P}_2$. When actual P_2 arrives at the same frontier, equality of the entire frozen input confirms j_3 under [EL]. Its execution continues with the same identity. Subscripts identify successive states, rather than audio seconds; audio, target, EOS, model and decoding bindings are also fixed.

The call can expose its own prediction $\widehat{P}_3$, yet original Correct/Update may choose a different P_3. [AU] permits selection of that complete original result once its parent is installed; it does not certify $\widehat{P}_3$. Independently, installing P_2 triggers [SUP] on the actual whole view h. A complete path from P_2 to h permits retention; otherwise the Consumer displays the installed state. Losing this path does not refute the unknown input of a later call. The protocol therefore distinguishes continuing computation, buffering a result, selecting a transition and retaining a view.

The following sufficient witnesses formalize these decisions. Producer failure changes message admission separately, as described in Section IV-E.

B. Complete dependency witnesses

The central question is whether an old call still belongs to a valid provisional continuation after trusted feedback advances. Its birth version records dispatch history. The dependency witness establishes whether it remains usable now.

Let $x = (F, P_C)$. A complete recorded node provides an edge

$$(F, P_C) \xrightarrow{n} (b_n, P_n^+) \quad \text{if} \quad a_n = F \ \wedge \ P_n^- = P_C, \tag{3}$$

under the input, identity, and execution contract of Section II. A finite sequence of such edges is a *valid provisional continuation* of C when each successor consumes the preceding complete output state at its actual frontier. Write $C \rightsquigarrow n$ for a complete path to n. A frozen in-flight successor can belong to the same continuation, but contributes no usable result until completion.

Proposition 1 (Sufficient re-rooting witness)**.** *Suppose an old branch contains a complete boundary node at the new certified frontier F, with output state P_C. If its subsequent recorded calls form an unbroken input–output path to the actual branch tip, those nodes are a valid provisional continuation of C. An in-flight successor may continue when its unchanged frozen input is that tip and its target exceeds F.*

Proof. The boundary equals (F, P_C), so the first successor satisfies Eq. (3). Each subsequent call consumes the preceding complete output at its recorded frontier under the same contract. Induction reaches the tip and the unchanged input of its in-flight successor, which supplies no output yet. □

This establishes dependency validity, not Canonical write authority. Provisional results remain revisable. The witness follows actual calls without assuming bitwise equality of different floating-point executions.

A branch born under v can reach (F, P_C) when $v+1$ installs, making version-only rejection wasteful. Conversely, equal text can hide different tokens or alignment state; equal state at a different frontier also fails Eq. (3).

C. [EL] Retaining computation: R

The core Qwen guard conservatively implements Proposition 1. On installation, it requires a complete Private node at exactly F with full feedback tokens P_C, then checks frozen inputs and frontiers along complete successors to the actual tip. An intermediate match is insufficient. An in-flight successor must be the unique tracked task, consume that tip, and target audio beyond F.

Success retains the branch without changing job IDs, birth versions, inputs, or results; otherwise the Owner fences it and rebuilds from an allowed state. Boundary-node and unique-slot requirements restrict this sufficient guard, not every valid reuse scheme.

D. [SUP] Retaining an actual view: S

A snapshot Consumer can replace a valid ahead view with a shorter certified snapshot despite computation retention. RevTrace maintains separate certified and provisional whole-hypothesis lanes, without an uncertified suffix splice or certified-prefix status for retained views.

For a Canonical application, S checks the complete visible node's input directly against (F, P_C) or follows complete published ancestors to that boundary. Missing, cyclic, or ambiguous ancestry is rejected. Unlike R's conservative guard, a directly rooted published call needs no extra Private boundary node.

Before actual application and its receipt, the kernel checks the preview against the durable root, registered results, dependency path, and bound text/frontier. A caller-supplied compatibility flag cannot replace these records.

Proposition 2 (Continuity at a compatible application)**.** *Immediately before an authorized Canonical application, let the visible whole provisional view h be ahead of its new frontier F. If S proves $C \rightsquigarrow h$ and no independent withdrawal mode is invoked, that application advances the certified lane and preserves h unchanged.*

Proof. The authorized compatible-and-ahead branch installs C in the certified lane while retaining h's node, text, and frontier. That application therefore withdraws none of h. □

Incompatible updates must withdraw the view, and later valid results can revise it. This is a per-application property, not global monotonicity. Snapshot consumption and conservative recomputation can be safe; the four-arm study measures the extra progress and cost of joint retention.

E. Native feedback and conditional full calls

StreamCorrect supplies a concrete recursive feedback task [1]. For complete closed state P and accepted audio X, expose its recognition boundary as

$$A(P, X) = (B, M), \tag{4}$$

where B contains ordered beam candidates and M the engine state and inputs needed by original Correct/Update. Candidate order, prompts, returned token IDs, selection, and update rules remain intact.

Before learned correction finishes, the original update tail with a typed first-beam identity prediction produces provisional closed state $\widehat{P}^{+}$. It is a dependency projection, not a complete selected result. A second worker can run the next *complete original call* on that conditional input; both GPUs host the full recognizer–corrector. Its own prediction may be wrong even when its input later matches. Original Correct/Update determines the complete result.

When the parent installs, the Owner compares the call's complete frozen start state, actual frontier, contract, and session epoch with it. Equality confirms the unchanged owned call: it continues if running, or becomes eligible for ordered selection if complete. Job ID, birth, producer incarnation, inputs, and execution history stay unchanged.

An earlier ancestor mismatch can remove current view support without refuting a later complete input. Cancellation before correction requires a disagreeing actual selected state at that call's own start frontier, coverage of its target without remaining use, or a changed epoch. An unresolved future parent supplies none of these proofs. The view can therefore require withdrawal while the call remains unresolved.

This native input witness and the core Qwen boundary-node/unique-successor guard are different sufficient implementations of the continuation relation. Neither grants write authority. Both use a whole-view Consumer that keeps certified and provisional provenance separate. The native comparison holds two complete workers fixed and varies conditional dispatch and retention; ready-only ASR handoff is a separate developmental control. For a call j, its complete frozen input is *confirmed* when it matches the installed parent, *refuted* when an actual selected state at its start frontier differs, and *unknown* without either witness. Covered targets and fenced sessions cannot supply selected results. Table I separates these input decisions from actual view application.

The dispatcher scans available workers and sessions in their existing order. Within each session it tries missing selected work before a conditional tail, subject to the ahead bound; it imposes no global selected-work priority across sessions. Unknown input grants neither selection nor a refutation-based cancellation proof. Replacing a typed prediction with its complete result keeps only the suffix that consumes the actual output contiguously. Physical release and producer-failure handling follow Section IV.

TABLE I
EVENT-DRIVEN COORDINATION OF CONFIRMED CALLS AND SUPPORTED WHOLE VIEWS.

Event	Required evidence	Action
Dispatch	Ready worker; accepted audio and an eligible session start	Use the unchanged dispatcher; freeze the full input, target and original job identity.
ASR result	Bound recognition result and contiguous provisional path	Extend the typed prediction and publish its provisional view; do not select it.
Full result	Bound original result [AU]; complete input status [EL]	Select if confirmed, buffer if unknown; deny use if refuted, covered or fenced.
Select ready	Confirmed complete result and unchanged installed parent	Synchronize the bound fact before installation; record the selected state and recheck pending calls.
Consumer update	Ordered fact, actual parent and whole-view support [SUP] $S(C, h)$	Advance the certified view. Retain the complete ahead h when supported; otherwise render the installed state.

F. [AU] Preserving the selected program

Let $P \Rightarrow_{\mathcal{A},X} P'$ denote a transition allowed by the original protected program on accepted immutable input X. The relation may be nondeterministic. A complete original call includes its original selection and update, or its declared buffer-only path; a typed identity projection is not such a result. We project a protocol execution onto its distinct durable selected facts, deleting speculation, waiting, local rejection, cancellation, collection, and repeated delivery. This restricted projection follows the standard stuttering view of refinement [7]; provisional whole-view behavior remains outside its alphabet and has a separate compatibility property.

Proposition 3 (Selected-trace refinement)**.** *Assume sufficient declared state, faithful execution of frozen inputs, exclusive serial selection, correctly bound message identities, and storage that honors successful synchronization. If each new durable fact binds a complete original call whose full input equals the preceding selected cut, retains the call's original identity, was admitted under valid message authority, and is synchronized before installation or certified application, then the projected facts form an execution allowed by $\mathcal{A}$. Replaying an existing fact adds no transition.*

Proof. The initial cut belongs to $\mathcal{A}$. Deleted actions do not extend the prefix. Each new fact binds a complete original call on exactly the preceding cut and accepted input, hence an allowed next edge regardless of dispatch time. Synchronization precedes privileged application. Recovery replays the same checked identities, including a fact first consumed after an Owner crash, without adding an edge. Induction proves the claim. □

The durable append is the selection boundary; installation and consumption do not select again. Already accepted buffers retain their producer identities after exit, whereas new fenced messages are inadmissible. The certified frontier derives only from checked facts; ahead views remain provisional. Input confirmation does not certify predicted next states.

A separately frozen specification explores 6,231 two-frontier normal states and 14,074 one-frontier failure states, each to a 12-event depth, with 27 named controls. An independent trace checker binds actual facts and whole-view applications to original call receipts. The specification is not extracted from the runtime; depth bounds truncate exploration, and neither check verifies all Python, CUDA, or operating-system behavior. Exact equality with a fixed neural reference is separate empirical evidence, rather than a premise that arbitrary GPU reruns must be bitwise identical.

IV. EXECUTION PROTOCOL

A. Complete results and physical ownership

In the core runtime, dispatch binds session, incarnation, job, input, target, and contract. A row returns either a complete result with re-derived state/text or cancellation without partial state. Complete rows can be installed and consumed before their batch finishes. Only all terminal rows and device synchronization authorize `BatchDrained` and executor reuse. The native process lifecycle below has separate evidence.

B. Covered work and safe-point exit: K

For the tested use model, K requests exit when the installed Canonical frontier covers a proposal's target and it has no remaining declared use. This differs from R: valid work beyond the new root can remain useful. Workers observe `CancelRequest` at decode boundaries; completion can win, but each row has one terminal outcome. Static batches are not compacted or refilled; masked costs and drain delays remain measured.

C. Durable facts and outstanding input

Before acknowledgment, synchronized frames bind parent/next states, request identity, selected complete result, and receipt; recorded deltas must reconstruct the state. Group-ready commit shares synchronization among validated sessions without a collection timeout or cross-session atomic transaction. Uncertain append/application outcomes block writes. Write-ahead recovery [8] reconstructs facts and identities; the native extension binds this to original correction calls.

D. Validation boundaries

Propositions 1 and 2 concern the stated guards and compatible applications. The artifact retains an earlier, separately frozen recovery prototype and its bounded model and crash results; they do not validate the later core or native fault paths. The selected-trace argument of Proposition 3, its finite specification, and actual execution checks have distinct scopes.

E. Continuing through delivery and process failures

Each job has an Owner namespace and monotonic issue number; each physical slot has a worker incarnation, using established RPC identity principles [9]. Fencing removes future message authority without renaming already accepted complete results.

a) Local rejection and physical release: The ingress guard rejects exact duplicates and fenced messages locally; they cannot mutate selected state or release newer leases. Contradictory currently owned contents freeze writes. Per-incarnation Pipes and receiver threads isolate incomplete frames. Replacement requires process exit, CUDA release and joined receivers. Strictly validate complete receipts from the sealed channel before fencing the incarnation and retiring unfinished work. Accepted buffers and supported views retain their original identities.

b) Durable selection and recovery: A synchronous hook binds complete parent/next states, original correction and IDs, job, Owner/session epochs, and producer incarnation. Framing, checksums and chaining protect the fact; synchronization precedes installation/publication. Uncertain writes freeze the writer. Recovery discards only an incomplete final frame; complete corruption fails. These conventional write-ahead rules [8] bind the original feedback transition.

The surviving Supervisor preserves the input ledger and original clocks. After old physical ownership ends, it advances the Owner namespace, replays the verified fact prefix, and cold-starts workers. Replay adds no inference; old publications gain no new authority. Both policies share this coordinator, durability, recovery, and actual Consumer.

c) Conditional continued service: For finite accepted work after finite supported failures, further progress requires readable storage, eventual physical release/loading, terminating original calls that advance frontiers, fair Owner/Consumer scheduling, and sufficient service budgets. This is not an RTO or D2/C8 guarantee; loading, replay and recomputation retain their original clocks.

d) Control history: Roots, calls, buffers, visible views and outboxes pin their dependencies. Actual acknowledgment releases publication references, while the visible view stays pinned. Namespace/issue high watermarks prevent reuse after collection. Unreferenced terminal and audit windows have fixed limits; diagnostics stream to disk. These bound protocol entries relative to live dependencies, separately from current state, cumulative text and total memory.

V. Evaluation

A. Design, comparators and service endpoints

We evaluate whether [EL], [AU] and [SUP] jointly preserve legal progress under unchanged selected feedback histories. Core factorial controls separate computation retention from whole-view retention; native tests measure confirmed full calls executing the original correction task. An instrumented FIFO Consumer records actual applications. All accepted inputs keep their original ready times through cancellation and recovery.

a) Core implementation and sources: Two 24-GB RTX 3090s host an FP16 Private worker and FP32 Authority worker. Qwen3-ASR 1.7B [10] is primary; Whisper base.en [11] tests a second state adapter. Complete-row delivery, at most 16 rows per batch, a 256-token limit, the protocol kernel and synchronized journals are shared. Physical early exit is disabled in this factorial study, separating retention from cancellation. Initial loading precedes the service clock. Energy integrates both physical boards over the declared service window.

Independent confirmation uses 24 previously unused LibriSpeech speakers [12], metadata-selected before inference and partitioned into three eight-speaker cohorts, totaling 714.110 s per arm. Its 110 cells comprise 60 primary cells (three cohorts, two backends, two arrival patterns, five arms), 20 fresh-process repeats on cohort A and 30 coarser-feedback controls. At $\rho = 1$, Authority advances every 2 s; $\rho = 3$ changes selection frequency and feedback segmentation. Arrivals are simultaneous or staggered by 0.25 s. Repetitions and arms reuse these sources; speakers are the independent acoustic units.

b) Named comparisons: Checked snapshot + recomputation (R_0S_0), *View retention only* (R_0S_1), *Computation retention only* (R_1S_0), and ***RevTrace: joint retention*** (R_1S_1) share validation, durability, input accounting and physical execution. Disabling R rebuilds work from selected feedback; disabling S replaces the provisional view with the installed snapshot while retaining the same dependency checks. The *Alternative joint implementation* is our complete positive control: another policy instantiation of the same retention and Consumer conditions on the shared execution/commit substrate. External prior work and our author-built comparators have separate identities.

c) Endpoints: For offered interval i, w_i is audio duration, r_i its original ready time, d_i first actual visible coverage and c_i first actual certified coverage. Audio coverage meeting both deadlines is

$$A_{2,8} = \sum_i w_i \, \mathbf{1}\{d_i - r_i \le 2 \ \wedge \ c_i - r_i \le 8\}. \qquad (5)$$

All offered intervals, including partial end-of-stream (EOS) tails, remain in the denominator. Whole-run passes require display P99 at most 2 s (D2) and certified-view P99 at most 8 s (C8), over offered frontiers. P99 weights frontier observations; $A_{2,8}$ weights duration, so a P99 pass may coexist with a small uncredited tail.

We separately count visible-frontier withdrawals, erased code points and deficit below the previously observed frontier. Retrospective whole-text support compares a first actual view with later selected text at that view's own frontier. This is distinct from dependency validity [SUP], human word/character error rate, and new character count. Independent checks reconstruct complete selected $F/Z/P/\mathrm{text}$ histories and actual consumed input paths from original outputs.

B. Does retaining computation retain delivered service?

Retaining computation alone can improve timeliness while increasing withdrawal. Figure 3(a) shows the predeclared Qwen, $\rho = 1$, staggered stratum. With the same selected histories, computation retention raises covered audio from 654.115 to 664.150 s but withdrawals from 12 to 17. View retention alone reduces withdrawals to 1 but covers 628.435 s. RevTrace jointly covers 680.760 s with zero frontier withdrawals: 26.645 s above recomputation and 16.610 s above computation-only.

Cohort-wise credit interaction, $I = Y_{11} - Y_{10} - Y_{01} + Y_{00}$, is $+18.530$, $+6.205$, and $+17.555$ s. These are three queue-coupled cohort contrasts, not independent deadline windows.

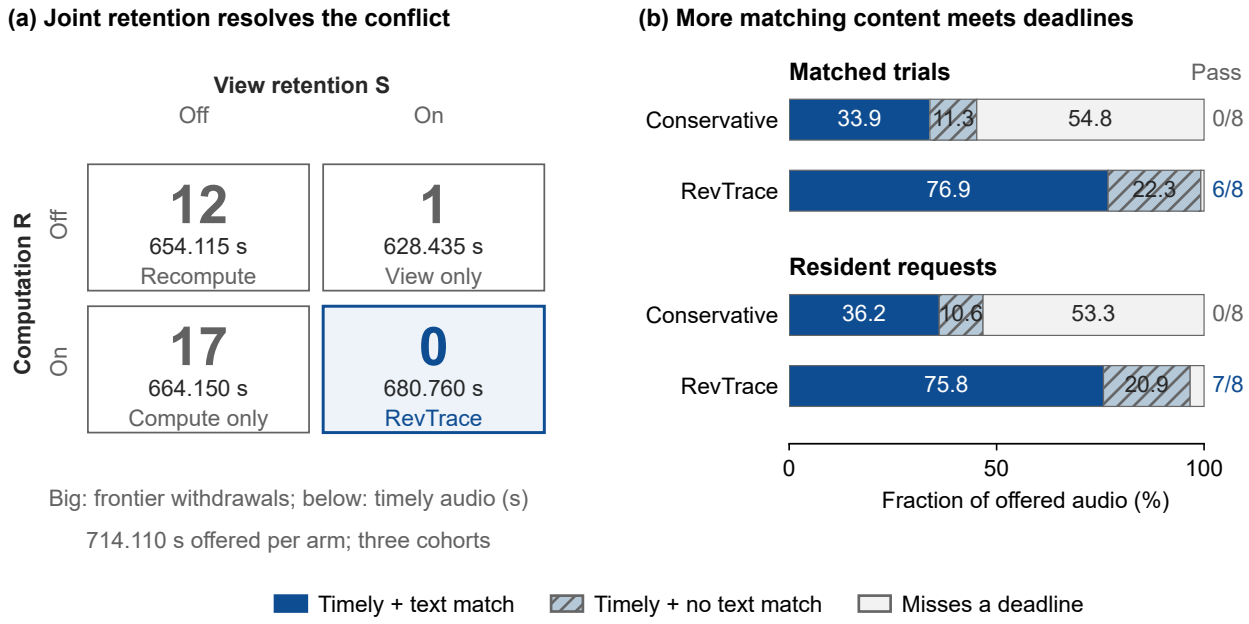


Fig. 3. Retained work and retained service require coordinated decisions. (a) The predeclared 24-speaker stratum: each cell sums three eight-speaker cohorts' frontier withdrawals and audio coverage meeting display 2 s/certified 8 s deadlines over 714.110 s per arm. The four matrix arms and our separate Alternative joint positive control (15 runs total) fail whole-run D2. (b) Final native matched and resident banks each offer 66.340 s per policy on four existing sources. Stacks separate timely same-frontier whole-text agreement, other timely coverage and uncovered audio; text agreement is retrospective against selected model output. Unlabelled tails are the remainder to 100%, with rounded labels. Passes count matched trials or resident requests meeting display P99 $\leq$ 2 s and certified P99 $\leq$ 8 s.

RevTrace's text remains revisable: 569 characters are erased despite zero frontier withdrawal. Median per-run display P99 is 2.329 s, above D2, so the benefit is retained progress under overload rather than a whole-run service pass.

The Alternative joint implementation also yields zero withdrawals and 569 erased characters, with 667.530 s coverage and 2.534-s median P99. It reproduces the supported-view behavior on the common substrate. Across confirmation, 144 matched history groups and 22 positive-control pairs agree on complete selected histories. Thus the comparison holds the feedback task fixed while changing how valid work becomes service.

Finding 1: retained computation does not ensure retained service. [EL] can retain work whose presentation a snapshot Consumer still removes. Joint [EL]/[SUP] coordination preserves the supported ahead view.

C. Timely native service with original full calls

Input-confirmed full calls improve timely service while preserving the original feedback task. In the final four-speaker comparison, RevTrace raises audio coverage meeting both deadlines from 45.2% to 99.2% and whole-run passes from zero to six of eight (Table II). The StreamCorrect adaptation retains original learned correction and feedback, using fixed 500-ms inputs, no VAD, and FP32 execution [1]. Both author-built policies have two complete workers, the same ASR early-view Consumer and neural warmup. Conservative execution waits for selected parent feedback; RevTrace starts conditional full calls and retains them when their complete inputs are confirmed.

TABLE II
FINAL NATIVE REVTRACE IMPROVES TIMELY COVERAGE WITH THE ORIGINAL SELECTED TASK. EACH POLICY OFFERS 66.340 AUDIO-S OVER FOUR PREVIOUSLY MEASURED SPEAKERS AND TWO FRESH TRIALS PER SPEAKER. COVERAGE AND BOTH-BOARD ENERGY ARE TOTALS; D99 IS THE MEDIAN OF PER-RUN DISPLAY P99. ENERGY SPANS THE COMMON EXTERNAL ORIGIN-TO-CONFIRMED-PHYSICAL-MODEL-DRAIN WINDOW. PASS COUNTS TRIALS MEETING BOTH WHOLE-RUN D2/C8 CRITERIA.

Policy	Timely coverage (s)	D99 (s)	Energy (kJ)	Pass
Conservative	30.000	4.759	47.019	0/8
RevTrace	65.822	1.638	43.952	6/8

a) Task fidelity and matched sources: The matched version study holds four AISHELL-1 sources [13], models, warmup, allocator and GPU-role rotation fixed: two versions, two policies and two fresh repeats give 32 runs. All 544 selected transitions match complete original inputs/outputs, correction and returned IDs under the declared static code-version mapping. The sources were initially selected by metadata in an 8–9-s band after freezing the earlier prototype. Its separate 16-run independent bank matched all 272 original transitions, with five character edits over 81 unique reference characters (6.17% CER) in both arms. The matched study reuses those speakers; repetitions do not add independent acoustic units. Every original 500-ms frontier and partial EOS

remains in the accounting.

b) Reliability-extension cost and EOS margin: Final RevTrace uses 6.52% less common-window board energy than final Conservative. Relative to the earlier matched RevTrace prototype, energy increases by 10.61% (39.735 to 43.952 kJ). That prototype achieves 8/8 passes and 66.340 s of coverage in the matched study; its earlier independent bank separately improves coverage from 37.0 to 66.33975 s and passes from 0/8 to 8/8. Final RevTrace's two remaining failures are S0908 EOS intervals, whose display P99 increases by 0.163/0.050 s against the earlier matched version. Other paired timing differences have both signs. The common model-drain window supports this energy comparison; source-specific cleanup and certified-EOS windows remain separate endpoints.

c) Resident independent requests: RevTrace raises joint D2/C8 passes from 0/8 to 7/8 requests, and timely coverage from 31.0 to 64.118875 of 66.33975 s. Models load once per trial; both worker PIDs persist across four pre-registered independent requests. The next request starts only after actual certified consumption of the previous file EOS. Two policies and two fresh trials preserve all 272 original transitions and each request's own input clock. The second S0908 repeat has D99 2.229 s. This single-outstanding closed-loop contract tests residency, rather than online VAD or open-arrival capacity.

d) The same content arrives earlier: Figure 3(b) separates deadline coverage from content support. Timely coverage whose whole first view retrospectively matches selected text at that view's own frontier G rises from 22.5 to 51.0 s in matched runs and from 24.0 to 50.259 s in resident requests. RevTrace's remaining 14.822/13.860 s of timely coverage does not match; unknown coverage is zero. Raw and whitespace comparisons agree. Each policy in either bank introduces 180 code-point occurrences, retracts 14 and leaves 166 last-visible. All-offered whole-text matches are 106/136 for Conservative and 104/136 for RevTrace. The gain is earlier delivery of matching content, with unchanged character counts; this comparison with selected model output is distinct from human-transcription accuracy.

e) Input confirmation and view retention: The earlier independent bank has 92 promoted calls whose recognition begins before selected parent completion; 20 have wrong own predicted ends. Twenty-four invalid/covered calls stop after ASR and two complete wrong-input results are discarded. These observations separate input eligibility from prediction correctness.

The final matched bank has no compatible-ahead application and therefore does not repeat the core factorial continuity test. Resident RevTrace has one nonempty opportunity: a 5.0-s whole view survives a compatible 4.5-s selected update with the same tip and job. Its 0.817-s Consumer event-state holding interval spans that update and ends at legitimate supersession. Each RevTrace bank also contains two necessary incompatible withdrawals; final objects are right-censored at the last actual same-source observation.

Finding 2: confirmed inputs accelerate the original task. Original selection [AU] remains intact when [EL] confirms early full calls; whole-view retention still requires [SUP] independently.

TABLE III
OPERATING BOUNDARIES REMAIN VISIBLE. EACH WORKLOAD HAS FOUR ARMS AND THREE FRESH PROCESSES (48 RUNS TOTAL). PASS COUNTS COVER ALL 12 RUNS; D99 RANGES SPAN THREE REVTRACE OBSERVATIONS, AND TIMELY-LOSS WINDOWS LIST COMPUTATION-ONLY/REVTRACE COUNTS. SOURCE MIX EXPANDS WITH SESSIONS. FULL 48-POINT PLOTS AND ALL ENDPOINTS REMAIN IN THE ARTIFACT.

Workload	D2/C8 pass	RevTrace D99 (s)	Timely-loss windows
8 sync	12/12	1.426–1.528	1,0,0 /0,0,0
8 staggered	12/12	1.704–1.955	1,2,1 /0,0,0
12 staggered	0/12	2.197–2.299	1,1,3 /0,0,0
16 staggered	0/12	2.801–3.223	10,12,9 /0,0,0

D. When does the opportunity appear or disappear?

The opportunity is a supported whole view ahead of a newly installed selected frontier. In the headline RevTrace arm, 18 of 368 updates encounter and retain one; computation-only encounters 17 and clears all 17. Under synchronized arrivals, RevTrace encounters only 3 of 368 and covers 0.790 s less than computation-only. A fresh staggered repeat on cohort A also changes the timing ranking: 226.320 versus 230.565 s, $I = -2.245$ s. Compatible relative progress determines the benefit; the three positive primary contrasts do not establish an invariant speed advantage.

Table III summarizes 36 predeclared development runs (8 synchronized, 8 staggered, 16 staggered sessions) and 12 exploratory midpoint runs (12 staggered). Each arm offers 121.205/185.300/250.180 s at 8/12/16 sessions. These are explicit workloads with different source mixes. At 8 sessions all arms meet whole-run D2/C8; RevTrace synchronized display P99 is 1.426–1.528 s and staggered reaches 1.955 s. Two 0.215-s tails remain uncredited despite their P99 passes. At 12 and 16 sessions all runs fail D2 and pass C8. At 16, RevTrace's extra coverage over computation-only is +18.910,+5.055,$-$7.225 s; P99 improves in one repeat. Its continuity distinction is stable: compatible withdrawals are zero versus 16/17/12.

A secondary analysis follows intervals already delivered within D2 whose eventual certified coverage also meets C8. At each selected application, the complete dependency path classifies compatible loss separately from necessary revision. Loss ends when actual coverage returns or is right-censored at the next selected root. Computation-only has 10/12/9 such windows at 16 sessions, with 24.903/23.635/15.736 audio-s$\times$wall-s deficit; RevTrace has none. Holes intersect the original D2 read point for 6.000/10.675/4.000 s of distinct audio. The 12-session strong witness is not established: computation-only's smaller losses never include a timely complete non-EOS interval under its frozen criterion.

Finding 3: continuity and first delivery have different boundaries. Retaining a supported view removes an avoidable

withdrawal; whether enough work arrives before D2 remains a separate scheduling and load outcome.

a) Quality and cost: Headline Qwen arms share WER 26/1908 (1.36%) and selected histories. Retrospective same-frontier whole-text agreement leaves their credits unchanged. Elsewhere RevTrace's raw comparison removes 2 s in the 16-session third repeat and 2 s in each midpoint's first two repeats, although the frozen normalizer agrees. Whisper WER is 83.12% at $\rho = 1$ versus 11.53% at $\rho = 3$; changing its feedback contract changes quality. Headline board energy spans 59.16–59.73 J/audio-s across five arm medians. At 16 sessions, RevTrace reduces total model time by 0.66–1.59 s over recomputation, while wall-time and energy differences change sign. Retained progress is the measured benefit; inference cost and overload are independent boundaries.

E. Preserving acquired results through process failures

The final coordinator preserves acquired results without granting old messages new authority. Both policies share models, durability, guards, recovery and Consumer. Initial loading and fixed warmup precede service; replacement loading, replay, recomputation and physical drain retain the original clock.

Six conditions, two policies and two fresh processes on one development clip give 24 short trials, plus four EOS/compatible-view regressions. Together with eight sustained-state trials below, these 36 trials contain 3,976 durable transitions independently bound to complete original call receipts and actual Consumer applications, with zero illegal installations or incorrect owned-message rejections. Eight duplicate/late messages cause no selected-state mutation; eight worker deaths have confirmed physical replacements. Conditional calls supply 354 durable facts: 352 have original Owner installation records, and two first reach the Consumer through recovery. In four worker-after-ASR cases, a previously consumed whole view remains across producer exit/fencing until legitimate correction. Its support, rather than the producer's continued presence, determines retention.

Worker and Owner failures preserve legality but have different service costs (Table IV). Normal and duplicate RevTrace each pass one of two trials; the other misses D2 only at EOS. All worker-fault trials fail D2 but pass C8; Owner-fault trials fail both. Old durable facts reappear in 1.36–1.81 s, whereas new certified post-fault completions take 64.61–108.08 s in Owner cases. A surviving worker can finish a call before replacement readiness, so replay, new results and replacement loading have separate recovery clocks.

a) Complete buffers across producer death: Two matching and two mismatching timing controls test the accepted-result case absent from the bank above. A transport fixture holds the parent after original ASR and the child after sending its complete original result. After the Owner registers the result, we kill its producer, verify exit, CUDA release, sealed-channel validation and fencing, then release original parent correction. Both matching results install and are consumed once with their original job, birth and incarnation; both mismatches are rejected against the actual same-frontier parent. Positive cases also finish a subsequent independent request. These controlled interleavings establish retention and rejection, rather than fault-time SLOs. Independent reconstruction corrects an audit-indexing defect over unchanged traces; the original failed audit gates remain in the artifact.

Finding 4: producer failure and result invalidity are different. Fencing ends future message authority while eligible, already accepted results retain their original identities and permitted uses.

b) Sustained-state stress and its limits: Two metadata-selected AISHELL-1 speakers contribute 24 utterances each, concatenated as exact PCM into 228.51025/208.1003125-s streams without join resets. A continuous original-engine reference and 875 separately restored steps agree on complete state, correction and IDs, crossing 398/357 buffer shifts and 353/252 context changes. Eight serving trials produce 3,500 matching selected steps. Unpinned terminal and each audit class stay at most 64 entries; live dependencies, state/output bytes, Owner RSS and cumulative logs are separate.

All eight trials fail D2/C8 (display P99 449.727–502.952 s). Both policies reproduce severe original-task insertion errors: CER 3103/486 (638.48%) and 1313/522 (251.53%). RevTrace uses more energy. The original length guard can skip decoding, so frontier coverage need not add text. These are state/collection stress tests rather than deployable continuous ASR. Across all 36 trials, the checker preserves 90 compatible-ahead dependency-tip applications and records 1,118 necessary incompatible withdrawals. Internal tips need not add text; these controlled counts do not estimate natural prevalence.

An independent dispatch audit finds 360 eventual matches among 1,901 conditional inputs; 1,541 are refuted, none unresolved. Six matched jobs never become durable after injected failures. Workload-specific ratios remain separate. Original-clock misses and a conservative offline excess-length bound precede the after-trim 448-token condition on each long stream; neither yields online prefix CER or a unique causal diagnosis.

F. From physical release to supported service

A separate 24-run core study crosses R and K with S enabled, two arrival patterns and three fresh processes, using a frozen wrong-conditioning fixture. Selected histories and WER $14/687$ match; all 675 same-input/prompt/audio/peer-order pairs return identical outputs. Synchronized $R1$ releases the next 16-row computation 473–577 ms earlier. All 96 following rows are actually consumed and their whole texts match later same-frontier Canonical text. Only 6/6/1 rows switch from deadline miss to hit. Adding K changes credit by +7.970 to +15.255 s when synchronized and -26.000 to $+21.780$ s when staggered, without consistent energy savings. Request acknowledgment still precedes physical drain. This is a measured release-to-service opportunity, with no stable positive $R \times K$ interaction.

TABLE IV

FAULT CONTAINMENT PRESERVES LEGALITY; COLD RECOVERY STILL MISSES THE ORIGINAL SERVICE DEADLINES. THE 24 SHORT TRIALS ALL FINISH WITH ZERO ILLEGAL INSTALLATIONS. EACH ROW COVERS TWO FRESH TRIALS ON THE SAME CLIP; REVTRACE USES INPUT-CONFIRMED FULL CALLS. REJECTED MESSAGES CAUSE ZERO SELECTED-STATE MUTATIONS (ΔC). RANGES ARE OBSERVED MINIMA/MAXIMA. NEW RESULT MEASURES FAULT TO ACTUAL CERTIFIED CONSUMPTION OF A NEW FACT WHOSE CALL COMPLETES AFTER THE FAULT, SEPARATELY FROM OLD-FACT REPLAY AND REPLACEMENT READINESS. PASS COUNTS JOINT WHOLE-RUN D2/C8; EOS COUNTS D2 MISSES. BOTH-BOARD ENERGY RUNS THROUGH PHYSICAL DRAIN.

Condition	Policy	Reject / ΔC	New result (s)	D2/C8 pass	EOS miss	Energy (J)
No fault	Conservative	0 / 0	–	0/2	2	6566–6574
	RevTrace	0 / 0	–	1/2	1	5896–5911
Duplicate / late	Conservative	4 / 0	–	0/2	2	5701–5893
	RevTrace	4 / 0	–	1/2	1	5538–5846
Worker after ASR	Conservative	0 / 0	1.90–2.22	0/2	2	5182–5401
	RevTrace	0 / 0	2.15–2.21	0/2	2	5447–5647
Worker partial frame	Conservative	0 / 0	2.10–2.25	0/2	2	4775–5178
	RevTrace	0 / 0	1.76–2.23	0/2	2	5492–5594
Owner before durable	Conservative	0 / 0	86.42–108.08	0/2	2	24549–28019
	RevTrace	0 / 0	64.61–107.09	0/2	2	19571–26557
Owner after durable	Conservative	0 / 0	83.90–87.05	0/2	2	22415–24200
	RevTrace	0 / 0	101.80–104.44	0/2	2	26483–27171

TABLE V

REPRESENTATIVE REUSE OBJECTS AND REVTRACE'S FEEDBACK-SPECIFIC OBLIGATION. ENTRIES DESCRIBE THE COMPARED MECHANISMS' FOCUS.

Family	Reuse object	Feedback obligation here
IU / Adapton	Units / dependency computations	Bind complete feedback inputs and owned calls.
SSBD / PredGen	Draft tokens / candidate replies	Preserve original recursive correction and update.
StreamCorrect	Selected correction prefix	Overlap full calls and validate the actual whole view.
Retire	Contract-valid successor state	Derive input and view actions from selected neural feedback.

VI. RELATED WORK

a) Incremental information and dependency reuse: The incremental-unit model separates newest committed information from the active right edge and propagates dependency revocation [2]. Its commit promises that a unit will not be revoked. Here, certification records acceptance by the protected selection protocol; later selected facts may still revise the transcript. Adapton checks recorded dependencies; Nominal Adapton combines names with arguments and dependency checks to recover reusable work [3], [4]. Differential Dataflow maintains incremental computations across evolving versions [14]. These works establish much of the basis for dependency-aware reuse (Table V). RevTrace instantiates that basis for complete neural feedback state and frozen executed calls. Its service obligation connects execution eligibility to the actual whole view retained when an independently selected root is applied. Our sufficient witness follows recorded executions; it does not transfer a from-scratch theorem to independently repeated floating-point inference.

b) Revisable speech services and speculation: Whisper-Streaming converts an offline recognizer into incremental service using LocalAgreement [15]. StreamCorrect adds learned correction and feeds selected corrections into subsequent recognition [1]. RevTrace retains that selection and feedback behavior in its controlled native adapter; its contribution concerns the asynchronous execution and consumption around the feedback boundary. It does not introduce a better acoustic model or train a new corrector.

SSBD coordinates old-output draft reuse with display-only suffix masking [5]. It revalidates a draft under updated input and includes an unbiased configuration. RevTrace follows complete calls that actually executed to establish their eligibility under newly selected feedback state, then checks the dependency witness for the Consumer's whole current view. PredGen provides a close precedent for speculation while speech input evolves across recognition, language generation, and speech synthesis [6]. Our comparison holds the complete selected feedback task fixed across the retention policies. The specific reuse object, protected selection, and whole-view transition define the composition evaluated here. RelayS2S uses a learned verifier to commit a fast-path prefix that constrains subsequent slow-path generation [16]. Our native calls instead confirm their complete inputs and retain the original correction and feedback update. Work on ASR alignment drift also establishes the importance and cost of audio position state [17]; our contract includes that state rather than treating text equality as sufficient.

c) Execution, authorization, and recovery: Retire jointly addresses revocation, valid state inheritance and physical reclamation for runtime-selected version transitions [18]. RevTrace derives complete-call eligibility and whole-view actions from an unchanged selected feedback task: an unsupported view can be withdrawn while a future call's input remains unresolved. Our covered-exit policy is execution support, without a new general cancellation claim. The parent-version control does not reproduce the complete Retire system.

Optimistic and multiversion concurrency control separate validation and commit from concurrent execution [19], [20]. Write-ahead recovery persists state changes and identities [8], and stream-processing systems coordinate recoverable state [21]. We use these established principles at the ASR

feedback boundary. They supply machinery for ordering and durability; the adapter still has to specify which state constitutes a valid continuation and which complete view the Consumer may retain.

VII. Discussion

a) Scope of the guarantee: The continuation witness assumes a sufficient adapter state and faithful execution of frozen inputs. A new adapter must account for every mutable field that can affect later inference, selection, or feedback. A hidden KV cache, alignment cursor, or external tool state can invalidate a token-only argument. Conservative rejection remains allowed when a sufficient witness is absent. The guarantee concerns dependency validity and an individual compatible application; it supplies neither transcription correctness nor monotone text for an entire session.

b) Progress, content, and capacity: Processing more audio can leave the cumulative transcript unchanged. Frontier credit, actual text additions and revisions, eventual selected text agreement, and ground-truth accuracy answer different questions. We report them with their original denominators. Preserving compatible progress also cannot remove all input synchronization, queueing, or neural work. The negative c12 witness identifies a timing boundary. An earlier native two-source test earns 17.0 s with retention versus 18.5 s for independent full replicas, at higher energy and a different warmup/Consumer. These do not define a universal capacity envelope. Independent native confirmation establishes a single-chain service opportunity with the original selected task; it does not turn that result into saturated multi-source capacity.

c) Deployment and validation: The continuous-original reference exercises the native cut across real buffer and context changes; another adapter must establish its own state sufficiency. Active-dependency collection bounds protocol entries relative to live references, separately from cumulative text, current state, and total memory. Owner/worker recovery assumes the surviving services of Section II. Replaying old facts restores access to that prefix; the measured cold Owner restarts still miss original deadlines for new work. Broader ownership transfer requires separate validation. Finite specifications, source controls, controlled faults, actual GPU service, and retrospective checks provide distinct evidence.

d) Eligibility and resource admission: A valid input witness determines whether an owned call may be used. It does not determine whether starting another conditional call is worthwhile. Limiting new speculation can leave existing supported whole views intact. The long stress study motivates this separate resource decision. In a bounded two-client probe, new speculative calls precede another client's ready necessary call in both repeats. The later client has no additional C8 misses, and its C99 penalty fails the predeclared repeat-consistency gate. No admission controller or general native-session fairness is evaluated.

VIII. Conclusion

Feedback revisions change which uses of speculative work remain valid. RevTrace coordinates complete-input eligibility, original selection authority and whole-view support so that a selected update can advance without an avoidable withdrawal of still-supported progress. The factorial study separates retained execution from retained service; native full calls improve timely coverage while preserving original correction and feedback. Process-failure controls distinguish accepted results that remain usable from newly fenced messages and refuted inputs. Observed EOS, load, quality and cold-recovery limits remain separate deployment obligations.

The practical lesson is to bind different uses to their own evidence: a complete frozen input for continuing a call, an authorized original result for selected advancement, and a whole dependency path for keeping a displayed view. Under the declared state and failure contract, these witnesses preserve useful service while allowing necessary correction.

AI Tool Usage

ChatGPT/Codex assisted literature discovery, drafting and revision of all manuscript sections and the refinement argument, runtime extensions and experiment orchestration, and validation/analysis code producing the figures and tables from recorded measurements. The artifact records execution sources, frozen configurations, controls, and result provenance, separating actual service measurements from component tests and retrospective analyses. Constructed streams concatenate recorded AISHELL-1 PCM. The authors retain responsibility for the submission.

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